Analytic Stress in DFT+eDMFT with LAPW

From Blaha's Eq. (13), through Eq. (20),
to a matrix-valued Appendix G2

This note derives the valence part of the stationary stress before using the diagonal Kohn–Sham occupation identities. It retains both the density matrix and its energy moment, makes the volume and representation cancellations explicit, and gives six local matrix contractions corresponding to the six lines of Blaha's Eq. (G2).
The normalized correlated projector is included in the functional derivative. The last sections identify the source routines to re-audit, record the scope of the available fixed-input tests, and distinguish derived identities from unresolved implementation questions.
Status: a derivation and audit reference. This is not a claim that the complete analytic eDMFT stress now agrees with the strain derivative of the free energy. No program source was changed to prepare this document.

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1 The result and how to read it

The starting point is the derivative of the stationary DFT+eDMFT functional, not the derivative of the usual total-energy estimator. At the stationary solution its valence, one-particle contribution is

$$D_\mu^{\mathrm{1p}} = \sum_{\mathbf{k}} w_{\mathbf{k}} \,\mathrm{Re}\,\mathrm{Tr}_b\big[\rho_{\mathbf{k}}\,\partial_\mu H_{\mathbf{k}} - \rho_{E\mathbf{k}}\,\partial_\mu O_{\mathbf{k}}\big]. \tag{1.1}$$

Here μ denotes a strain channel, H is the KS Hamiltonian matrix, O is the LAPW overlap matrix in the chosen representation, and ρ_E is the energy-weighted density matrix. Stress is D_μ/Ω. The full derivative also contains the normalized-projector derivative, the remaining density-functional and ion terms, and any explicitly strain-dependent interaction parameters.

In a fixed central-band representation the valence contribution can be organized as

$$D_\mu^{\mathrm{1p}} = \sum_{\mathbf{k}} w_{\mathbf{k}} \,\mathrm{Re}\,\mathrm{Tr}_b\big[\rho(T^\mu + V^{\mu,\mathrm{op}} + B^\mu) - \rho_E O^{\mu,\eta}\big]. \tag{1.2}$$

T^μ is the explicit kinetic-operator strain derivative; $V^{\mu,\mathrm{op}}$ is the explicit potential-operator derivative in material coordinates; B^μ and $O^{\mu,\eta}$ contain the volume-compensated basis derivative. Their integral definitions are given in Sec. 10. The superscript η distinguishes this overlap contribution from a derivative in an arbitrary moving representation.

The principal generalization of Eq. (20). The scalar DFT factor $f_n\epsilon_n$ must become the matrix ρ_E. It cannot in general be replaced by $\frac{1}{2}\{\mathcal{E},\rho\}$. The full Hamiltonian and overlap bilinear forms must be retained before contraction.

Three points keep this statement from being mistaken for an ad hoc correction.

1. The normalization part of the first term in Blaha's Eq. (13) cancels the volume Jacobian. It is not a separate $\mathrm{Tr}\,M_\Sigma$ pressure term.
2. A change of band representation contributes to the one-particle expression, but its pure representation part cancels against the corresponding projector change. A changing truncated subspace is a different matter.
3. The six expressions derived here describe the frozen-augmentation, nonrelativistic valence formula. A complete RELA implementation, radial-solution response, and the final total-stress assembly still require independent checks.

Sections 3–5 give the functional and geometrical derivation. Sections 7–9 expand the result into local radial and angular integrals. Sections 11 and 14 connect the mathematics to the next implementation audit.

2 Notation, units, and what is held fixed

2.1 Indices and matrix spaces

$\mathbf{k}, w_{\mathbf{k}}$	Brillouin-zone point and its fixed integration weight; $\partial_\mu w_{\mathbf{k}} = 0$. A spin block can be included in the index $\mathbf{k}$; spin multiplicity is counted once, not again in the formulas.
m, n	Bands in the chosen valence space; this space may include the noninteracting bands outside the DMFT window.
K	LAPW basis index, including local-orbital basis functions when present.
a	Atom in the unit cell, with position τ_a and muffin-tin radius R_a.
$L = (l, m_l)$	Spherical-harmonic index. m_l is a magnetic quantum number, distinct from a band index m.

$i = (L, \lambda), j = (L', \lambda')$	Local augmentation indices. For the elementary LAPW pair, $\lambda = 0$ denotes u_l and $\lambda = 1$ denotes $\dot{u}_l = \partial u_l / \partial E$.
ℓ	Correlated-orbital index, including the correlated site and spin structure as appropriate.
$\mu = (\alpha\beta)$	One of six symmetric strain channels; $\alpha, \beta \in \{x, y, z\}$.
Ω	Unstrained cell volume.
β_T, T	Inverse temperature and temperature in energy units; $T = 1/\beta_T$. We use β_T to avoid confusion with a Cartesian index.
ω_j, z_j	Fermionic frequency $\omega_j = (2j+1)\pi T$ and $z_j = \mathrm{i}\omega_j + \mu_{\mathrm{chem}}$.

The band matrices H, O, G, ρ, ρ_E have size $N_b \times N_b$. At the unstrained reference $O = \mathbb{I}$ and $H = \mathcal{E} = \mathrm{diag}(\epsilon_1, \ldots, \epsilon_{N_b})$. The local coefficient matrices A, D^μ have size $N_{\mathrm{aug}} \times N_b$. The normalized projector U and its raw counterpart P have size $N_b \times N_c$; the gradient C has size $N_c \times N_b$. Local matrices S^x, H^x, V^x have size $N_{\mathrm{aug}} \times N_{\mathrm{aug}}$. Tr_b and Tr_c denote band and correlated-orbital traces.

We define $\mathrm{Herm}\, X = (X + X^\dagger)/2$, the anticommutator $\{A, B\} = AB + BA$, and the commutator $[A, B] = AB - BA$. The elementary trace convention is $\mathrm{Tr}_b(\rho X) = \sum_{mn} \rho_{nm} X_{mn}$. Unless shown explicitly, all matrices in a product have the same $\mathbf{k}$ and spin block. No k weight is absorbed into ρ in this note.

2.2 Strain and stress conventions

Let $\boldsymbol{\varepsilon}$ be a symmetric strain tensor and define

$$F = \mathbb{I} + \boldsymbol{\varepsilon}, \qquad \mathbf{r}' = F\mathbf{r}, \qquad J = \det F, \qquad \Omega' = J\Omega. \tag{2.1}$$

For a channel $\mu = (\alpha\beta)$ use the generator

$$S^\mu_{ij} = \tfrac{1}{2}(\delta_{i\alpha}\delta_{j\beta} + \delta_{i\beta}\delta_{j\alpha}), \quad \partial_\mu F = S^\mu, \quad \gamma_\mu = \partial_\mu \ln J = \delta_{\alpha\beta}. \tag{2.2}$$

Thus the off-diagonal channel parameter is the engineering shear: $F_{\alpha\beta} = F_{\beta\alpha} = s/2$. If a test instead sets both off-diagonal entries equal to s, its derivative is twice the channel derivative used here.

The thermodynamic sign convention is

$$\sigma_\mu = \Omega^{-1} \partial_\mu \Gamma_{\mathrm{stat}}, \qquad p = -\tfrac{1}{3} \mathrm{Tr}\, \boldsymbol{\sigma}. \tag{2.3}$$

For isotropic linear strain $F = (1+h)\mathbb{I}$, $\mathrm{d}\Gamma/\mathrm{d}h = \Omega(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$. Divide by 3Ω when comparing the hydrostatic mean stress.

Equations using $H = -\nabla^2/2 + V$ are in **Hartree atomic units**, as in the paper. The code uses **Rydberg** energies: multiply Hamiltonians, energy moments, and energy derivatives by two, while keeping overlaps unchanged. In particular the inhomogeneous energy-derivative radial equation must use the same energy variable as the radial function. Approximately $1\,\mathrm{Ry}/a_0^3 = 14710.5\,\mathrm{GPa}$.

2.3 Scope of the path

The elementary derivation holds R_a fixed in bohr, moves atomic centers affinely, holds local axes fixed, and uses a fixed number of basis functions and fixed band-index window. Radial functions are frozen as functions of their physical radius and energy parameter, but their coordinate arguments are strained. The distinction is crucial: frozen augmentation does *not* remove the $ru'(r)$ terms.

A finite-difference comparison must specify what is held fixed in the potential, radial functions, projector, charge, self-energy, chemical potential, and band window. A fixed-input matrix test and a relaxed free-energy test are different derivatives.

3 Stationary functional and the two density matrices

3.1 Which functional is differentiated?

A useful auxiliary-field form of the DFT+eDMFT stationary functional is

$$\Gamma = \mathcal{L}[H_{\mathrm{KS}}, O, U, \widetilde{\Sigma}] - \int_{\Omega} \mathrm{d}^3 r \, [V_{\mathrm{KS}} - V_{\mathrm{ext}}] \, n + E_{\mathrm{Hxc}}[n] + E_{\mathrm{ion}} + \Phi_{\mathrm{imp}}[G_{\mathrm{loc}}] - \Phi_{\mathrm{DC}}[n_{\mathrm{loc}}] - \mathrm{Tr}_{\omega,c}(\widetilde{\Sigma} G_{\mathrm{loc}}) + \mu_{\mathrm{chem}} N. \tag{3.1}$$

The last term converts the grand potential to the fixed-N Helmholtz functional; omit it when working at fixed chemical potential. $n(\mathbf{r})$ is the charge density, V_{ext} the external ionic potential, E_{Hxc} the Hartree plus exchange-correlation functional, and E_{ion} the ionic electrostatic energy. Φ_{imp} is the local many-body functional and Φ_{DC} its double-counting subtraction. The local self-energy used for upfolding is $\widetilde{\Sigma} = \Sigma_{\mathrm{imp}} - V_{\mathrm{DC}}$. The trace $\mathrm{Tr}_{\omega,c}$ includes temperature and all Matsubara frequencies as well as local orbital/site traces.

$\mathcal{L}$ is the *regularized* lattice trace-log contribution associated with

$$G_{\mathbf{k}}(z)^{-1} = z O_{\mathbf{k}} - H_{\mathbf{k}} - U_{\mathbf{k}} \widetilde{\Sigma}(z) U_{\mathbf{k}}^{\dagger}. \tag{3.2}$$

It represents $-T \sum_{\mathbf{k}j} w_{\mathbf{k}} \ln \det[-G_{\mathbf{k}}^{-1}(z_j)]$ with the high-frequency reference and basis-measure normalization included. The raw unnormalized determinant is not a definition of the thermodynamic functional in a nonorthogonal basis. Only its finite, consistently regularized derivative is needed below.

At stationarity, variations of the auxiliary fields and local Green function cancel. In particular $\delta(\Phi_{\mathrm{imp}} - \Phi_{\mathrm{DC}}) = \mathrm{Tr}_{\omega,c}(\widetilde{\Sigma} \delta G_{\mathrm{loc}})$ cancels the corresponding term in $-\delta \, \mathrm{Tr}_{\omega,c}(\widetilde{\Sigma} G_{\mathrm{loc}})$; the lattice trace-log supplies the opposite $\mathrm{Tr}_{\omega,c}(G_{\mathrm{loc}} \delta \widetilde{\Sigma})$. The explicit KS-potential variation cancels through the density-functional part. This statement does not discard changes of U induced by a changing KS potential, band subspace, or radial functions: these remain in the specified total projector derivative. No impurity two-particle vertex is required for this first derivative at stationarity.

Equation (3.1) is a stationary functional, not a claim that every frozen-self-energy path is stationary. A numerical frozen-Σ check must use the agreed linear impurity-functional correction and a consistent charge-relaxation path.

3.2 Density and energy moment

At the central orthonormal KS representation define

$$\rho_{\mathbf{k}} = T \sum_{j}^{\mathrm{reg}} G_{\mathbf{k}}(z_j), \qquad \rho_{E\mathbf{k}} = T \sum_{j}^{\mathrm{reg}} z_j G_{\mathbf{k}}(z_j). \tag{3.3}$$

The superscript "reg" includes the convergence prescription, the high-frequency contact subtraction, and analytically summed reference. It must be identical to that used in the trace log. It is fixed so that for $\widetilde{\Sigma} = 0$,

$$\rho_{mn} = f_n \delta_{mn}, \qquad (\rho_E)_{mn} = f_n \epsilon_n \delta_{mn}, \tag{3.4}$$

where f_n is the Fermi occupation at the chosen temperature. Both matrices in Eq. (3.3) are Hermitian after the complete frequency sum.

The finite derivative of the lattice term is

$$\delta\mathcal{L} = \sum_{\mathbf{k}} w_{\mathbf{k}} \, \mathrm{Re} \, \mathrm{Tr}_b \left(\rho \delta H - \rho_E \delta O + T \sum_{j}^{\mathrm{reg}} G(z_j) \, \delta\Sigma^{\mathrm{emb}}(z_j) \right), \tag{3.5}$$

with $\Sigma^{\mathrm{emb}} = U \widetilde{\Sigma} U^{\dagger}$. The part involving $\delta\widetilde{\Sigma}$ cancels in Eq. (3.1); the explicit δU part remains. This gives Eq. (1.1) and the projector term derived in Sec. 11.

3.3 What is M_Σ?

Left and right Dyson identities give

$$B_L = \rho_E - \mathcal{E}\rho = T\sum_j^{\text{reg}} \Sigma^{\text{emb}}(z_j)G(z_j), \tag{3.6}$$

$$B_R = \rho_E - \rho\mathcal{E} = B_L^\dagger, \tag{3.7}$$

$$M_\Sigma = \tfrac{1}{2}(B_L + B_R) = \rho_E - \tfrac{1}{2}\{\mathcal{E}, \rho\}. \tag{3.8}$$

The symbol B_L is a Dyson residual moment, not the basis derivative matrix B^μ introduced later. Explicitly,

$$M_\Sigma = \frac{T}{2}\sum_j^{\text{reg}} [\Sigma^{\text{emb}}G + G\Sigma^{\text{emb}}]. \tag{3.9}$$

M_Σ has energy units and vanishes in the noninteracting limit with the same Hamiltonian reference. The decomposition

$$\rho_E = \tfrac{1}{2}\{\mathcal{E}, \rho\} + M_\Sigma \tag{3.10}$$

is exact. Replacing ρ_E by a symmetrized band energy omits M_Σ and is not justified by merely Hermitizing the DFT formula.

4 Recovering Blaha's Eq. (13) without discarding a term

Let $\phi_K[\varepsilon](\mathbf{r}')$ be a strained LAPW basis function and $c_{Kn}[\varepsilon]$ a coefficient. The material-coordinate wave function is

$$\widetilde{\psi}_n[\varepsilon](\mathbf{r}) = \psi_n[\varepsilon](F\mathbf{r}) = \sum_K c_{Kn}[\varepsilon]\, \phi_K[\varepsilon](F\mathbf{r}). \tag{4.1}$$

All derivatives below are evaluated at zero strain; a tilde is suppressed after this definition. Define three derivatives:

$$b_n^\mu(\mathbf{r}) = \sum_K (\partial_\mu c_{Kn})\phi_K(\mathbf{r}), \tag{4.2}$$

$$\eta_n^\mu(\mathbf{r}) = \sum_K \frac{c_{Kn}}{\sqrt{\Omega}}\partial_\mu \left[\sqrt{\Omega[\varepsilon]}\phi_K[\varepsilon](F\mathbf{r})\right]_0, \tag{4.3}$$

$$\zeta_n^\mu = b_n^\mu - \frac{\gamma_\mu}{2}\psi_n. \tag{4.4}$$

The product rule now gives exactly the separation in the paper:

$$\boxed{\partial_\mu \psi_n = b_n^\mu - \frac{\gamma_\mu}{2}\psi_n + \eta_n^\mu = \zeta_n^\mu + \eta_n^\mu.} \tag{4.5}$$

Blaha's first term is ζ_n^μ, not just b_n^μ. It combines the derivative of the expansion coefficients and the derivative of $\Omega^{-1/2}$. His second term is η_n^μ.

4.1 Why the first term vanishes in the DFT derivation

In a finite LAPW basis the KS eigenproblem implies

$$\langle \phi_K | H_{\text{KS}} - \epsilon_n | \psi_n \rangle = 0 \quad \text{for every } K. \tag{4.6}$$

This is a projected variational equation; it does not say that the finite-basis wave function solves the continuum differential equation pointwise. Since ζ_n^μ lies in the same basis space, $\langle \zeta_n^\mu | H - \epsilon_n | \psi_n \rangle = 0$. That is the precise reason for the sentence below Eq. (13) in the paper.

In DMFT the relevant contraction contains ρ and ρ_E, not a separate factor $f_n(H-\epsilon_n)$ for each state. One must therefore return to the matrix derivative before making this simplification. The next section does so, including the Jacobian that is already canceled in the DFT Rayleigh quotient leading to the paper's Eq. (12).

4.2 Interstitial behavior

For an interstitial plane wave,

$$\sqrt{\Omega[\varepsilon]}\phi_{\mathbf{k}K}[\varepsilon](F\mathbf{r}) = \exp[\mathrm{i}F^{-T}(\mathbf{k}+\mathbf{K})\cdot F\mathbf{r}] = \exp[\mathrm{i}(\mathbf{k}+\mathbf{K})\cdot\mathbf{r}]. \tag{4.7}$$

Hence $\eta_n^\mu = 0$ there. This is the content of Appendix D1. It does *not* set the interstitial kinetic stress to zero: the kinetic operator itself changes under the pullback.

5 The volume cancellation and generalized Eq. (20)

5.1 Differentiate the matrix integrals first

Write $\dot\psi_n = \partial_\mu\psi_n$ and let $H^{\mu,\mathrm{op}}$ denote the derivative of the pulled-back differential operator, with the wave functions held fixed. Differentiating the whole-cell matrix integrals gives

$$(\partial_\mu H)_{mn} = \gamma_\mu H_{mn} + \langle\dot\psi_m|H|\psi_n\rangle + \langle\psi_m|H|\dot\psi_n\rangle + \langle\psi_m|H^{\mu,\mathrm{op}}|\psi_n\rangle, \tag{5.1}$$

$$(\partial_\mu O)_{mn} = \gamma_\mu O_{mn} + \langle\dot\psi_m|\psi_n\rangle + \langle\psi_m|\dot\psi_n\rangle. \tag{5.2}$$

These formulas use the complete periodic cell. If a calculation instead differentiates separate moving MT/interstitial domains, their boundary terms must reconstruct these identities. The elementary formulas assume continuous LAPW functions with the required matching; APW derivative jumps and associated surface terms need separate care.

For a set of variations x_n define the real linear contraction

$$\mathcal{P}_\mu[x] = 2\,\mathrm{Re}\sum_{mn}\left[\rho_{nm}\langle x_m|H|\psi_n\rangle - (\rho_E)_{nm}\langle x_m|\psi_n\rangle\right]. \tag{5.3}$$

At one $\mathbf{k}$, Eq. (1.1) becomes

$$D_\mu^{1\mathrm{p}} = \gamma_\mu\,\mathrm{Tr}_b(\rho H - \rho_E O) + \mathcal{P}_\mu[\dot\psi] + \mathrm{Re}\,\mathrm{Tr}_b(\rho H^{\mu,\mathrm{op}}). \tag{5.4}$$

Hermiticity makes the displayed traces real where appropriate.

Insert Eq. (4.5). Its normalization contribution satisfies

$$\mathcal{P}_\mu[-\gamma_\mu\psi/2] = -\gamma_\mu\,\mathrm{Tr}_b(\rho H - \rho_E O). \tag{5.5}$$

Therefore

$$\boxed{D_\mu^{1\mathrm{p}} = \mathrm{Re}\,\mathrm{Tr}_b(\rho H^{\mu,\mathrm{op}}) + \mathcal{P}_\mu[\eta] + \mathcal{P}_\mu[b].} \tag{5.6}$$

The cancellation is valid also when $[\rho,\mathcal{E}]\neq 0$ and $M_\Sigma \neq 0$. At the central orthonormal reference, $\mathrm{Tr}(\rho H - \rho_E) = -\,\mathrm{Tr}\,M_\Sigma$. Restoring only the $-\gamma\psi/2$ piece would produce $+\gamma\,\mathrm{Tr}\,M_\Sigma$; the omitted Jacobian is exactly its negative.

No standalone volume correction follows from Eq. (13). Keep the Jacobian and wave-function normalization together. Their cancellation is a whole-cell identity and must precede restricting η to muffin-tin integrals.

5.2 The matrix-valued valence correction

With fixed central coefficients, $b = 0$. Since $\eta = 0$ in the interstitial, the generalized counterpart of the paper's Eq. (20) is

$$\boxed{\sigma_\mu^{\rm corr} = \frac{2}{\Omega}\,\mathrm{Re}\sum_{a\mathbf{k}} w_{\mathbf{k}} \sum_{mn} \left[\rho_{nm}\langle\eta_m^\mu|H_{\rm KS}|\psi_n\rangle_a - (\rho_E)_{nm}\langle\eta_m^\mu|\psi_n\rangle_a\right].} \tag{5.7}$$

The subscript a means integration over the central MT sphere of atom a. For Eq. (3.4), this reduces to

$$\sigma_\mu^{\rm corr,DFT} = \frac{2}{\Omega}\,\mathrm{Re}\sum_{a\mathbf{k}n} w_{\mathbf{k}} f_n \langle\eta_n^\mu|H_{\rm KS} - \epsilon_n|\psi_n\rangle_a, \tag{5.8}$$

which is precisely the structure of Eq. (20).

The potential in the matrix element in Eq. (5.7) is part of the basis derivative and must remain there. Cancellation of the *induced potential variation* in the stationary functional is not permission to delete this potential from the Pulay matrix element.

6 Band representation versus physical subspace response

Suppose a part of the coefficient response can be expressed inside the chosen band span as

$$b_n = \sum_m \psi_m L_{mn}. \tag{6.1}$$

Then the term retained in Eq. (5.6) is

$$\begin{aligned}\mathcal{P}_\mu[b] &= 2\,\mathrm{Re}\,\mathrm{Tr}_b[L^\dagger(\mathcal{E}\rho - \rho_E)] \\ &= 2\,\mathrm{Re}\,\mathrm{Tr}_b\Big[L^\dagger\left(\tfrac{1}{2}[\mathcal{E},\rho] - M_\Sigma\right)\Big] = -2\,\mathrm{Re}\,\mathrm{Tr}_b(L^\dagger B_L).\end{aligned} \tag{6.2}$$

This vanishes in DFT. It generally does not vanish in the DMFT one-particle term alone. That fact does not establish a new physical stress, because the same change of representation also changes U.

Let

$$C = T\sum_j^{\rm reg} \tilde{\Sigma}(z_j)U^\dagger G(z_j), \qquad UC = B_L. \tag{6.3}$$

For a pure orthonormal change of band frame, $L^\dagger = -L$ and $\delta U = L^\dagger U$. Its projector contribution is

$$\delta D^{\rm proj} = 2\,\mathrm{Re}\,\mathrm{Tr}_c(C\delta U) = 2\,\mathrm{Re}\,\mathrm{Tr}_b(L^\dagger UC) = 2\,\mathrm{Re}\,\mathrm{Tr}_b(L^\dagger B_L), \tag{6.4}$$

which cancels Eq. (6.2). The same covariance holds for a nonorthogonal frame when its overlap metric and projector normalization are transformed consistently; see Sec. 11.

6.1 Two consistent ways to organize the calculation

1. **Fixed reference coefficients.** Use the central coefficients to build ∂H and ∂O as basis derivatives. Then $b = 0$ in these matrix derivatives. Compute the projector derivative in the same representation, including metric factors.
2. **A moving KS basis.** Retain its coefficient response in both the one-particle and projector derivatives. Remove the common pure-frame contribution only after demonstrating the cancellation.

Mixing the first convention for one part and the second for another can leave a large, unphysical residual even when the separate matrix derivatives are accurate.

For a fixed-index window $W_{\mathbf{k}}$, rotation entirely within $W_{\mathbf{k}}$ is a change of frame. Mixing with states outside $W_{\mathbf{k}}$ changes the physical subspace selected by the window and cannot be discarded by the preceding argument. A constant number of bands does not imply $\partial_\mu \Pi_{W_{\mathbf{k}}} = 0$, where $\Pi_{W_{\mathbf{k}}}$ is its projector. If the window cuts a degeneracy, this derivative may not be uniquely smooth; include the degenerate multiplet or otherwise define a smooth subspace.

7 Local form of the basis derivative

7.1 Augmentation coefficients and radial functions

Within atom a use $\mathbf{r}_a = \mathbf{r} - \tau_a$, $r = |\mathbf{r}_a|$, and $\hat{\mathbf{r}} = \mathbf{r}_a/r$. Write

$$f_i^a(\mathbf{r}_a) = u_i^a(r) Y_{L_i}(\hat{\mathbf{r}}), \qquad \psi_n(\mathbf{r}_a) = \sum_i A_{in}^a f_i^a(\mathbf{r}_a), \qquad A_{in}^a = \sum_K c_{Kn} a_{iK}^a. \tag{7.1}$$

a_{iK} are the LAPW matching coefficients. The symbol A here is Blaha's local augmentation coefficient, not the full LAPW eigenvector c. For the basic LAPW pair

$$u_{l0} = u_l(r, E_l), \qquad u_{l1} = \dot{u}_l(r, E_l), \qquad \dot{u}_l = \partial u_l/\partial E. \tag{7.2}$$

A prime always means $\mathrm{d}/\mathrm{d}r$; a dot means an energy derivative. The spherical-harmonic normalization is $\int \mathrm{d}\hat{\mathbf{r}}\, Y_L^* Y_{L'} = \delta_{LL'}$.

Define the volume-compensated matching derivative at fixed c:

$$D_{in}^{a\mu} = \sum_K \frac{c_{Kn}}{\sqrt{\Omega}} \partial_\mu [\sqrt{\Omega[\varepsilon]}\, a_{iK}^a[\varepsilon]]_0. \tag{7.3}$$

This is Blaha's G3–G4. If s_{in}^μ is the *physical* fixed-c derivative of A_{in}, including $\Omega^{-1/2}$, then

$$D^\mu = s^\mu + \tfrac{1}{2}\gamma_\mu A. \tag{7.4}$$

One must not identify a routine returning s^μ with a routine returning D^μ without this conversion.

7.2 Radial and angular coordinate derivatives

Define the scalar angular multiplier and angular differential operator

$$p_\mu(\hat{\mathbf{r}}) = \hat{r}_\alpha \hat{r}_\beta, \qquad \mathscr{A}_\mu = \tfrac{1}{2}(\hat{r}_\beta \nabla_{\Omega,\alpha} + \hat{r}_\alpha \nabla_{\Omega,\beta}). \tag{7.5}$$

$\boldsymbol{\nabla}_\Omega$ is the dimensionless tangential gradient on the unit sphere. Since $\boldsymbol{\nabla} = \hat{\mathbf{r}}\,\partial_r + r^{-1}\boldsymbol{\nabla}_\Omega$,

$$(S^\mu \mathbf{r}_a) \cdot \boldsymbol{\nabla}[u_i(r) Y_i(\hat{\mathbf{r}})] = r u_i'(r) p_\mu Y_i + u_i(r) \mathscr{A}_\mu Y_i. \tag{7.6}$$

Here Y_i abbreviates Y_{L_i}. For isotropic dilation, summing the three diagonal channels gives $\sum_\alpha p_{\alpha\alpha} = 1$ and $\sum_\alpha \mathscr{A}_{\alpha\alpha} = 0$.

Combining the matching and coordinate derivatives gives

$$\boxed{\eta_n^\mu = \sum_i \left[D_{in}^\mu u_i Y_i + A_{in}\, r u_i' p_\mu Y_i + A_{in}\, u_i \mathscr{A}_\mu Y_i \right].} \tag{7.7}$$

These are the three channels that produce the six terms in G2 when $H_{\mathrm{KS}} = H_{\mathrm{sph}} + V_{\mathrm{ns}}$ is used.

7.3 What is omitted by frozen augmentation?

An explicit response of the radial solution would add

$$\eta_n^{\mu,\text{rad resp}} = \sum_i A_{in}(\partial_\mu u_i)_{r\,\text{fixed}} Y_i, \tag{7.8}$$

together with the consistent response of matching coefficients, linearization energies, local orbitals, and normalized projector. This is distinct from the ru_i' term already in Eq. (7.7). The present note does not establish that this omitted response is the source of the remaining total-stress discrepancy.

8 Local integral matrices: definitions without diagonal assumptions

Suppress a, μ temporarily and introduce three bra functions:

$$g_i^0 = u_i Y_i, \qquad g_i^r = r u_i' p_\mu Y_i, \qquad g_i^{\text{ang}} = u_i \mathscr{A}_\mu Y_i. \tag{8.1}$$

For $x \in \{0, r, \text{ang}\}$ define

$$S_{ij}^x = \int_{r<R_a} \mathrm{d}^3 r\, (g_i^x)^* f_j, \tag{8.2}$$

$$H_{ij}^x = \int_{r<R_a} \mathrm{d}^3 r\, (g_i^x)^* H_{\text{sph}} f_j, \tag{8.3}$$

$$V_{ij}^x = \int_{r<R_a} \mathrm{d}^3 r\, (g_i^x)^* V_{\text{ns}} f_j. \tag{8.4}$$

These are *one-sided* bilinear forms, with the differential Hamiltonian acting on the ket. They need not individually be Hermitian. The final band matrices are Hermitian sums, not individually symmetrized local factors. Integration by parts changes the radial volume matrix and introduces a surface term; both must be changed together.

8.1 Overlap integrals

The three local overlaps are

$$S_{ij}^0 = \delta_{L_i L_j} \int_0^{R_a} r^2 \mathrm{d}r\, u_i^* u_j, \tag{8.5}$$

$$S_{ij}^r = \left[\int_0^{R_a} r^3 \mathrm{d}r\, u_i'^* u_j\right] \int \mathrm{d}\hat{\mathbf{r}}\, Y_i^* p_\mu Y_j, \tag{8.6}$$

$$S_{ij}^{\text{ang}} = \left[\int_0^{R_a} r^2 \mathrm{d}r\, u_i^* u_j\right] \int \mathrm{d}\hat{\mathbf{r}}\, (\mathscr{A}_\mu Y_i)^* Y_j. \tag{8.7}$$

The angular derivative can connect different l values. The radial integral in Eq. (8.7) must therefore be evaluated for its actual pair (l_i, l_j). The familiar same-l identities $\langle u_l | u_l \rangle = 1$ and $\langle u_l | \dot{u}_l \rangle = 0$ cannot be applied across different l.

8.2 Spherical Hamiltonian action

In Hartree units the radial equations imply

$$H_{msph}(u_l Y_L) = E_l u_l Y_L, \qquad H_{msph}(\dot{u}_l Y_L) = E_l \dot{u}_l Y_L + u_l Y_L. \tag{8.8}$$

For any channel x,

$$\boxed{H^x_{ij} = E_{l_j} S^x_{ij} + \delta_{\lambda_j,1} S^x_{i,j_0},} \tag{8.9}$$

where $j_0 = (L_j, 0)$. No band eigenvalue has entered this local identity. It is therefore safe to use before taking an interacting density-matrix trace. Actual radial-normalization and energy-unit conventions must match Eq. (8.8).

8.3 Nonspherical potential integrals

Write

$$V_{\rm ns}(\mathbf{r}) = \sum_{L''\neq 0} V_{L''}(r) Y_{L''}(\hat{\mathbf{r}}). \tag{8.10}$$

Then

$$V^0_{ij} = \sum_{L''\neq 0} \left[\int_0^{R_a} r^2 \mathrm{d}r\, u_i^* V_{L''} u_j\right] \int \mathrm{d}\hat{\mathbf{r}}\, Y_i^* Y_{L''} Y_j, \tag{8.11}$$

$$V^r_{ij} = \sum_{L''\neq 0} \left[\int_0^{R_a} r^3 \mathrm{d}r\, u_i'^* V_{L''} u_j\right] \int \mathrm{d}\hat{\mathbf{r}}\, Y_i^* p_\mu Y_{L''} Y_j, \tag{8.12}$$

$$V^{\rm ang}_{ij} = \sum_{L''\neq 0} \left[\int_0^{R_a} r^2 \mathrm{d}r\, u_i^* V_{L''} u_j\right] \int \mathrm{d}\hat{\mathbf{r}}\, (\mathscr{A}_\mu Y_i)^* Y_{L''} Y_j. \tag{8.13}$$

Expanding p_μ and $\mathscr{A}_\mu Y_i$ in spherical harmonics gives the Gaunt-coefficient sums of G2. The integral form fixes all conjugations and derivative placements without depending on a particular Gaunt table.

9 The six generalized lines of Appendix G2

Insert Eq. (7.7) into Eq. (5.7), split $H = H_{\rm sph} + V_{\rm ns}$, and use the local matrices above. For one atom and one $\mathbf{k}$, define the six real energy derivatives:

$$d^\mu_1 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho D^{\mu\dagger} H^0 A - \rho_E D^{\mu\dagger} S^0 A], \tag{9.1}$$

$$d^\mu_2 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho A^\dagger H^r A - \rho_E A^\dagger S^r A], \tag{9.2}$$

$$d^\mu_3 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho A^\dagger H^{\rm ang} A - \rho_E A^\dagger S^{\rm ang} A], \tag{9.3}$$

$$d^\mu_4 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho D^{\mu\dagger} V^0 A], \tag{9.4}$$

$$d^\mu_5 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho A^\dagger V^r A], \tag{9.5}$$

$$d^\mu_6 = 2\,\mathrm{Re}\,\mathrm{Tr}_b[\rho A^\dagger V^{\rm ang} A]. \tag{9.6}$$

Their sum is

$$\boxed{\sigma^{\rm corr}_\mu = \frac{1}{\Omega} \sum_{a\mathbf{k}} w_{\mathbf{k}} \sum_{q=1}^{6} d_q^{a\mathbf{k},\mu}.} \tag{9.7}$$

The H^x, S^x, V^x for $x = r, \mathrm{ang}$ depend on μ through their angular functions even though this superscript is suppressed.

G2 line	Basis derivative	Local operator / overlap content
1	Matching, D^μ	H^0 and S^0
2	Radial coordinate, ru'	H^r and S^r
3	Angular coordinate	$H^{\rm ang}$ and $S^{\rm ang}$
4	Matching, D^μ	Nonspherical potential V^0
5	Radial coordinate, ru'	Nonspherical potential V^r
6	Angular coordinate	Nonspherical potential $V^{\rm ang}$

9.1 An explicit index expression

The first line, after using the radial equation, is

$$d_1^\mu = 2\,\mathrm{Re}\sum_{mnij}(D_{im}^\mu)^* A_{jn}\Big[(E_{l_j}\rho_{nm} - (\rho_E)_{nm})S_{ij}^0 + \rho_{nm}\delta_{\lambda_j,1}S_{i,j_0}^0\Big]. \tag{9.8}$$

The second and third lines have exactly the same bracket, with D_{im}^μ replaced by A_{im} and S^0 replaced by S^r or S^{ang}. In DFT, $\rho_{nm} = f_n\delta_{nm}$ and $(\rho_E)_{nm} = f_n\epsilon_n\delta_{nm}$, so the bracket contains $f_n(E_{l_j} - \epsilon_n)$ and recovers the corresponding G2 factors.

The off-diagonal generalization therefore does require changing one band index on the bra relative to the ket, as suggested in the discussion. It also requires replacing the entire energy-weighted occupation by ρ_E, including M_Σ. The latter information cannot be recovered from occupations alone.

9.2 Where the symmetric combination belongs

For any Hermitian ρ and any band matrix X,

$$2\,\mathrm{Re}\,\mathrm{Tr}_b(\rho X) = \mathrm{Tr}_b[\rho(X + X^\dagger)] = 2\,\mathrm{Tr}_b(\rho\,\mathrm{Herm}\,X). \tag{9.9}$$

For example, the Hermitian matching-potential band matrix is

$$D^\dagger V^0 A + A^\dagger (V^0)^\dagger D. \tag{9.10}$$

In components this produces the paired $D_m^* A_n$ and $A_m^* D_n$ terms. Using $\mathrm{Herm}\,X = (X + X^\dagger)/2$ is equally correct if the external factor two is retained. Applying a half and then dropping that factor loses half the contribution. Hermitize the *complete band matrix*, not a local radial factor whose bra and ket play different roles.

10 Explicit band matrices T^μ, B^μ, and O^μ

Define the one-sided band matrices

$$Y^\mu = \sum_a \left[D^{a\mu\dagger} S^{a0} A^a + A^{a\dagger} S^{ar} A^a + A^{a\dagger} S^{a,\mathrm{ang}} A^a\right], \tag{10.1}$$

$$Z^\mu = \sum_a \Big[D^{a\mu\dagger}(H^{a0} + V^{a0})A^a + A^{a\dagger}(H^{ar} + V^{ar})A^a + A^{a\dagger}(H^{a,\mathrm{ang}} + V^{a,\mathrm{ang}})A^a\Big]. \tag{10.2}$$

Then

$$\boxed{O^{\mu,\eta} = Y^\mu + Y^{\mu\dagger}, \qquad B^\mu = Z^\mu + Z^{\mu\dagger}.} \tag{10.3}$$

Equation (9.7) is equivalently

$$\sigma_\mu^{\mathrm{corr}} = \frac{1}{\Omega}\sum_{\mathbf{k}} w_{\mathbf{k}}\,\mathrm{Re}\,\mathrm{Tr}_b[\rho B^\mu - \rho_E O^{\mu,\eta}]. \tag{10.4}$$

These are definitions of the full off-diagonal matrices, not just their diagonal entries. They identify which local information is needed even if an implementation contracts the matrices without storing them.

For the explicit kinetic operator,

$$T^{\mu}_{mn} = \frac{1}{2}\int_{\Omega} \mathrm{d}^3 r\, \psi^*_m(\partial_\alpha\partial_\beta + \partial_\beta\partial_\alpha)\psi_n. \tag{10.5}$$

This is the off-diagonal bilinear form underlying Eqs. (19) and (G1). For smooth periodic functions, integration by parts gives the equivalent form

$$T^{\mu}_{mn} = -\frac{1}{2}\int_{\Omega} \mathrm{d}^3 r\, [(\partial_\alpha\psi_m)^*(\partial_\beta\psi_n) + (\partial_\beta\psi_m)^*(\partial_\alpha\psi_n)]. \tag{10.6}$$

When splitting the domain or using derivative-discontinuous basis functions, keep the corresponding surface contributions consistently.

The explicit potential-operator matrix is

$$V^{\mu,\mathrm{op}}_{mn} = \int_{\Omega} \mathrm{d}^3 r\, \psi^*_m(\mathbf{r})\partial_\mu[V_{\mathrm{KS}}[\varepsilon](F\mathbf{r})]_0\psi_n(\mathbf{r}). \tag{10.7}$$

In the fixed-c, whole-cell convention, within the specified basis approximation,

$$\boxed{\partial_\mu O = O^{\mu,\eta}, \qquad \partial_\mu H = T^\mu + V^{\mu,\mathrm{op}} + B^\mu.} \tag{10.8}$$

A moving representation adds $L^\dagger + L$ to ∂O and $L^\dagger\mathcal{E} + \mathcal{E}L$ to ∂H. These additions lead to Eq. (6.2).

10.1 What an energy-linear "kernel" means

A diagonal DFT routine can encode a bilinear form as

$$K^\mu(E) = T^\mu + B^\mu - EO^{\mu,\eta}. \tag{10.9}$$

Here "kernel" means a matrix representing this quadratic/bilinear form; it is not a two-particle vertex. If Eq. (10.9) has been established for the *full* matrix, then in a chosen energy unit $K^\mu(0) - K^\mu(1) = O^{\mu,\eta}$. Knowing only $\langle n|K^\mu(\epsilon_n)|n\rangle$ does not prove that this subtraction reconstructs the off-diagonal overlap derivative. Rearrangements valid on KS eigenstates may remove information needed for that inference. The integral definitions above provide the direct audit target.

11 Normalized projectors and consistency of the representation

11.1 The projector is not just a raw overlap

For raw projected local orbitals φ_ℓ, define in an orthonormal window basis

$$P_{n\ell}(\mathbf{k}) = \langle\psi_{n\mathbf{k}}|\varphi_\ell\rangle, \quad \mathcal{S} = \sum_{\mathbf{k}} w_{\mathbf{k}} P^\dagger_{\mathbf{k}} P_{\mathbf{k}}, \quad X = \mathcal{S}^{-1/2}, \quad U_{\mathbf{k}} = P_{\mathbf{k}} X. \tag{11.1}$$

This notation represents one normalization block; site/symmetry blocks must be treated as in the actual projector construction. The metric $\mathcal{S}$ is an orbital-space matrix and must not be confused with the band overlap O or the local integrals S^x. When symmetry is used after the k sum, the same symmetry operation must be applied to its derivative. Fixed local frames make those rotation matrices fixed; moving frames add their derivatives.

For fixed weights,

$$\partial_\mu U_{\mathbf{k}} = (\partial_\mu P_{\mathbf{k}})X + P_{\mathbf{k}}\partial_\mu X, \tag{11.2}$$

$$\partial_\mu \mathcal{S} = \sum_{\mathbf{k}} w_{\mathbf{k}}[(\partial_\mu P_{\mathbf{k}})^\dagger P_{\mathbf{k}} + P^\dagger_{\mathbf{k}}\partial_\mu P_{\mathbf{k}}]. \tag{11.3}$$

Writing $Y = \mathcal{S}^{1/2} = X^{-1}$, differentiating $X\mathcal{S}X = \mathbb{I}$ gives the Sylvester equation

$$Y\partial_\mu X + (\partial_\mu X)Y = -X(\partial_\mu \mathcal{S})X. \tag{11.4}$$

In an eigenbasis of $\mathcal{S}$ with positive eigenvalues s_i,

$$(\partial_\mu X)_{ij} = -\frac{(\partial_\mu \mathcal{S})_{ij}}{\sqrt{s_i}\sqrt{s_j}(\sqrt{s_i}+\sqrt{s_j})}. \tag{11.5}$$

This is well behaved for coincident positive eigenvalues. A singular global normalization block requires a specified rank treatment; differentiability can fail when that rank changes.

11.2 An essential qualification for a fixed-coefficient frame

The temporary strained states obtained by holding central LAPW coefficients fixed are not, in general, orthonormal. In that frame the same physical normalization is

$$\mathcal{S} = \sum_{\mathbf{k}} w_{\mathbf{k}} P_{\mathbf{k}}^\dagger O_{W,\mathbf{k}}^{-1} P_{\mathbf{k}}, \tag{11.6}$$

where O_W is the overlap within the chosen window span and P is the covariant overlap with its basis functions. At the central point $O_W = \mathbb{I}$,

$$\partial_\mu \mathcal{S} = \sum_{\mathbf{k}} w_{\mathbf{k}} \left[(\partial_\mu P)^\dagger P + P^\dagger \partial_\mu P - P^\dagger (\partial_\mu O_W) P \right]. \tag{11.7}$$

Equations (11.3) and (11.7) express the same normalization in different frames. The last term in Eq. (11.7) is *not* an extra term to add to a correct derivative already evaluated in a moving orthonormal KS basis.

For a pure frame transformation $\psi \mapsto \psi(\mathbb{I} + L)$, $\delta P = L^\dagger P$ and $\delta O = L^\dagger + L$ at the reference. Equation (11.7) then gives $\delta\mathcal{S} = 0$. Consequently $\delta X = 0$, $\delta U = L^\dagger U$, and the cancellation in Eq. (6.4) holds also for this nonorthogonal relabeling. This is a useful representation-consistency test, not a claim that the current code has omitted this term.

11.3 Projector stress and the direct gradient

The remaining explicit projector contribution is

$$\boxed{D_\mu^{\mathrm{proj}} = 2\,\mathrm{Re}\sum_{\mathbf{k}} w_{\mathbf{k}}\,\mathrm{Tr}_c[C_{\mathbf{k}}\partial_\mu U_{\mathbf{k}}], \qquad C_{\mathbf{k}} = T\sum_j^{\mathrm{reg}} \widetilde{\Sigma}(z_j) U_{\mathbf{k}}^\dagger G_{\mathbf{k}}(z_j).} \tag{11.8}$$

There is no dagger between C and ∂U: their rectangular orientations already make the product an orbital matrix. The frequency sum has the same regularization as the density moments.

One can infer C from $UC = B_L$ using $C = (U^\dagger U)^+ U^\dagger B_L$, where $+$ denotes a Moore–Penrose pseudoinverse. If U has deficient column rank at one $\mathbf{k}$, this reconstructs only the part of C visible through multiplication by U. The Matsubara expression in Eq. (11.8) defines the direct gradient without such an inversion. Agreement of contracted gradients is a test; it is not a replacement for the derivative of X.

12 The projector boundary and other extensions

12.1 A finite projector at the MT boundary

Suppose a projector is represented as $\varphi_\ell(\mathbf{r}) = \theta(R_a - r_a)\, q_\ell(\mathbf{r}_a)$, where q_ℓ does not vanish at R_a. A moving-domain overlap obeys the transport formula

$$\delta \int_{B_a} \mathrm{d}^3 r\, \psi^* q = \int_{B_a} \mathrm{d}^3 r\, \delta(\psi^* q) + \int_{\partial B_a} \mathrm{d}S\, v_n \psi^* q. \tag{12.1}$$

Here B_a is the integration ball and v_n is the boundary's normal displacement per unit perturbation in the coordinates being used. Equivalently, at fixed coordinate $\mathbf{r}$,

$$\delta\theta(R_a - r_a) = \delta(R_a - r_a)[\delta R_a + \hat{\mathbf{r}}_a \cdot \delta\tau_a]. \tag{12.2}$$

For fixed-radius physical spheres, $\delta R_a = 0$ but the sphere translates with its center. In atom-centered physical coordinates that translation is transferred into the derivative of the wave function; there is no additional translating-boundary term to add a second time. In a material pullback the same fixed physical sphere becomes a deformed integration region, so its shape derivative must be handled.

The fact that $q(R_a) \neq 0$ means a boundary contribution cannot be discarded merely by declaring the projector localized. It does not prove a missing term in an existing radial/matching implementation: the same contribution may already be contained in its coordinate transformation. A direct overlap derivative in a specified coordinate system is the decisive audit. This projector boundary issue is distinct from basis continuity used in deriving Eq. (5.1).

12.2 Local orbitals

Local orbitals can be included by enlarging the radial primitive set and the matrices A, D, S, H, V. For an extra primitive $u_l(r, E_{\mathrm{LO}})$, $H_{\mathrm{sph}} u_l = E_{\mathrm{LO}} u_l$. For a linear combination defining an LO, apply the Hamiltonian to every primitive and differentiate every coefficient and normalization constraint. Do not reuse the elementary $\delta_{\lambda,1}$ term for an arbitrary LO without deriving its radial action. APW+lo also requires its own boundary bookkeeping.

12.3 Scalar relativity and local-axis constraints

The cancellation of Jacobian, basis representation, and projector normalization is representation algebra and survives RELA. Equations (8.8) and (10.5), however, use the nonrelativistic differential operator. A RELA extension requires the actual scalar-relativistic radial Hamiltonian, small-component metric, and their strain derivatives. Successful RELA fixed-input matrix tests do not by themselves establish every term in the physical strain path.

If the self-energy is restricted to be diagonal in local correlated axes, changing those axes changes the approximation. A co-moving frame requires the derivative of its orbital rotation in P, U and the corresponding treatment of the imposed self-energy constraint. The present formulas use fixed local axes. Merely rotating a diagonal self-energy after truncation is not equivalent to a fully rotationally covariant calculation.

13 What comes directly from the paper, and what is new here

Paper location	Role in this derivation
Eq. (10)	DFT Rayleigh quotient. Its normalization cancels the volume factor before Eq. (12); it cannot be reused unchanged with a general interacting energy moment.
Eq. (11)	Explicit nonrelativistic operator derivative, including the kinetic expression in Eq. (10.5).
Eq. (13)	Product-rule decomposition. Rewritten exactly as $b - \gamma\psi/2 + \eta$ in Eq. (4.5).
Eq. (20)	DFT incomplete-basis correction. Generalized to independent ρ and ρ_E in Eq. (5.7).
Appendix D	Volume-compensated interstitial invariance and local radial/angular differentiation.
Eq. (G1)	Diagonal-occupation expansion of the kinetic bilinear form. Its off-diagonal definition is Eq. (10.5).
Eq. (G2)	Six diagonal-occupation local contractions. Their matrix versions are Eqs. (9.1)–(9.6).
Eqs. (G3–G4)	Matching coefficient derivative D, including the compensating $\sqrt{\Omega}$ factor.

Blaha's paper does not state the interacting ρ_E, M_Σ, normalized-projector gradient, or the matrix formulas in the notation used here. Those are the generalizations derived in this note. The diagonal DFT result is a necessary limit; it does not by itself determine all off-diagonal matrix elements.

13.1 Two places where the integral definitions are safer

In the supplied PDF, the definition of the radial quantity W below G4 prints radial derivatives on both radial functions. The direct derivation of G2 line 5, Eq. (8.12), instead gives

$$W_{i,L'',j} = \int_0^{R_a} r^3 \mathrm{d}r\, u_i'^*(r) V_{L''}(r) u_j(r), \tag{13.1}$$

with a derivative only on the bra. This follows because only the bra wave function is differentiated in the one-sided term; its conjugate is supplied by the outer $2\,\mathrm{Re}$. The printed two-prime definition appears to be a typographical error. The inspected `src/dftKS/atpar.f90` already implements and documents the single-prime form, so this observation is not a newly discovered code bug.

The third line of G2 also uses compact same-channel radial norm notation. When its angular operator connects different l, use the cross-l overlap of Eq. (8.7), rather than extending same-l orthogonality identities to that case. The inspected stress-integral code already stores such cross-l integrals. This is an audit of the meaning of the printed simplification, not evidence that those integrals are absent from the implementation.

13.2 Beyond simply changing one band index

The radial and angular operator structures in G2 need no new DMFT-specific angular algebra. The essential changes are (i) full bilinear band matrices, (ii) the independent energy moment, (iii) consistent volume and band-frame cancellation, and (iv) the derivative of the normalized correlated projector. Additional radial response, boundary handling, and relativity are separate approximations to audit.

14 Implementation map and the next audit

The paths below are relative to `/Users/haule/projects/eDMFT_new/` and refer to the inspected working tree on 6 November 2026. They locate relevant code; they do not certify that its total assembly is correct.

Source location	Mathematical responsibility / audit question
`src/dmft2/cmp_dmft_stress.f90`	Contracts density and energy-moment information with the existing stress kernels. Check the complete off-diagonal bilinear form against $T^\mu, B^\mu, O^{\mu,\eta}$, not only its DFT diagonal.
`cmp_dmft_paired_matching_matrix` in the same file	Diagnostic fixed-input matching derivatives, with separate spherical, surface, nonspherical, and interstitial pieces. These form a distinct derivative convention that must be mapped to the material-coordinate formula before substitution.
`src/dmft2/alm_blm_clm.f90`, `cmp_alm_blm_clm_strain`	Matching derivatives include physical volume-normalization terms. Determine explicitly whether each consumer needs s^μ or Blaha's $D^\mu = s^\mu + \gamma_\mu A/2$.
`src/dmft2/stress_integrals.f90`	Radial derivative and overlap integrals, including cross-l J and I channels. Compare with Eqs. (8.6) and (8.7), including small-component conventions.
`src/dftKS/atpar.f90`	Nonspherical V and single-prime W radial integrals; confirm all bra/ket and LO channels.
`src/dmft2/l2main.f90`	Construction and assembly of diagnostic density/moment contractions. Keep complete total-stress assembly separate from a successful fixed-input derivative check.
`src/dmft2/stress_modules.f90`	Diagnostic switches, including `EDMFT_STRESS_PAIRED_DIAG` and `EDMFT_STRESS_PAIRED_ASSEMBLY`. A diagnostic reconstruction is not a validated production replacement.

14.1 A concrete order for re-auditing

1. **Fix the derivative convention in writing.** For every routine record its coordinates, physical sphere motion, normalization, c treatment, potential held fixed, and Hartree/Ry and shear conventions.
2. **Construct local integral reference matrices.** Compare all six G2 channels with Eqs. (9.1)–(9.6) before assembling the final stress. Test both real and imaginary off-diagonal Hermitian density matrices; real test vectors alone cannot expose every complex off-diagonal error.
3. **Recover full H' and O' in one frame.** Check Eq. (10.8), including whole-cell volume cancellation, MT surfaces, interstitial terms, and all LO contributions. Do not infer O' from an energy-linear kernel without establishing its full bilinear identity.
4. **Close the projector derivative in the same frame.** Include X', window response, and the metric form of normalization when appropriate. Verify $UC = B_L$ and compare the direct gradient against pseudoinverse reconstruction. Check projector boundary motion through a direct overlap derivative.
5. **Verify exact invariances.** A pure band-frame rotation must cancel between Eqs. (6.2) and (6.4); a fixed-coefficient test has no coefficient-response term. In the noninteracting limit $M_\Sigma = 0$

and the DFT result must return.

6. **Only then assemble the physical functional derivative.** Keep the KS field-response cancellation and density-functional terms consistent with the chosen path. Compare NREL first, then RELA, with the prescribed frozen-Σ reference functional and relaxed charge. Use at least two strain magnitudes and test a shear or nonhydrostatic channel as well as isotropic strain.

The present evidence supports deriving and testing the complete off-diagonal bilinear forms. It does not support adding an isolated $M_\Sigma O'$ contribution to a total stress whose overlap and projector bookkeeping has not yet been shown to use the same convention.

15 What the recent fixed-input tests establish

The recent diagnostic contracts the central interacting moments with Hamiltonian and overlap derivatives formed on a specified fixed-input path. Define

$$\rho_o = \rho - \operatorname{diag}\rho, \qquad D_\mu^{\mathrm{extra}} = \operatorname{Re}\operatorname{Tr}_b\left[\rho_o H'_\mu - \left(\tfrac{1}{2}\{\mathcal{E},\rho_o\} + M_\Sigma\right) O'_\mu\right]. \tag{15.1}$$

The k sum and weights are implicit in this equation only. The quantity called "extra" is a diagnostic difference from a chosen diagonal reference. It is not the total DMFT stress and not an independently justified addition to the production result. The recorded paired diagnostic uses the raw valence-stress assembly sign, opposite to the energy-derivative sign of this table; for example, the MnO tag `PAIRED_MAIN_CORRECTION_COMPLETE_MEAN_GPA` is −101.143097512, corresponding to +101.143097512 here.

For isotropic linear strain $h = 0.001$, the recorded hydrostatic means in the physical energy-derivative sign convention are:

Material	Analytic	Central difference	Analytic − FD
	GPa, $D^{\mathrm{extra}}/(3\Omega)$		
$SrVO_3$	−5.701366	−5.695318	−0.006048
FeSe, RELA	17.599107	17.600406	−0.001299
MnO	101.143098	101.101274	0.041823

The decomposition of the finite-difference contractions was

Material	$\rho_o H' - \frac{1}{2}\{\mathcal{E},\rho_o\}O'$	$-M_\Sigma O'$
	GPa after k sum and division by 3Ω	
$SrVO_3$	−8.066304	2.370986
FeSe, RELA	−1.690087	19.290493
MnO	−11.585019	112.686293

These values were obtained before preparation of this PDF; no electronic structure calculation was rerun for this documentation task. The large MnO and FeSe components demonstrate why an isolated moment term is not a reliable physical correction. A complete derivative may contain large canceling terms.

Established within these tests: the reported analytic fixed-input contraction is close to the corresponding central matrix difference in these hydrostatic examples.

Not established: convergence with strain step, all individual matrix elements, shear channels, the full normalized-projector derivative in the same frame, or agreement of the complete stationary

free-energy derivative. One-sided derivatives show appreciable curvature in some cases. The MnO diagnostic process also reported IEEE invalid/divide-by-zero flags at exit; its printed contractions were finite, but the flags were not traced. This remains a diagnostic limitation rather than a clean numerical-validation claim.

15.1 Recorded locations for reproducibility

The checker is `src/python/check_oneparticle_stress_fd.py`. The following paths identify the diagnostic snapshots used; temporary directories are not a durable archive and may later be removed.

- $SrVO_3$: center and moments under `/private/tmp/sr_fixedpot_center_case/fd/`; displaced matrices under `/private/tmp/sr_fixed_joint_minus.LMwYDZ/fd/` and `/private/tmp/sr_fixed_joint_plus.J6UBtd/fd/`. Reference volume $382.70904\,a_0^3$.
- FeSe RELA: center and moments under `/private/tmp/fese_rela_fullq.QNEvK6/center/fd/`; displaced matrices under `/private/tmp/edmft_fixedpot_oracle_20261106/fese_minus/fd/` and the corresponding `fese_plus/fd/` directory. Reference volume $528.96251\,a_0^3$.
- MnO: center and moments under `/private/tmp/mno_projector_closure.hEJUBK/center/fd/`; displaced matrices under `/private/tmp/edmft_fixedpot_oracle_20261106/mno_minus/fd/` and the corresponding `mno_plus/fd/` directory. Reference volume $148.11716\,a_0^3$. Analytic diagnostic output: `/private/tmp/mno_paired_review.y3ZYV4/MnO.outputdmf2`.

16 Compact formula sheet for the next discussion

With all quantities in one representation and the same regularization,

$$D_\mu^{\mathrm{1p}} = \sum_{\mathbf{k}} w_{\mathbf{k}} \,\mathrm{Re}\,\mathrm{Tr}_b(\rho\partial_\mu H - \rho_E \partial_\mu O), \tag{16.1}$$

$$\rho_E = \tfrac{1}{2}\{\mathcal{E},\rho\} + M_\Sigma, \tag{16.2}$$

$$\partial_\mu \psi = b - \tfrac{1}{2}\gamma_\mu \psi + \eta, \tag{16.3}$$

$$\gamma_\mu \,\mathrm{Tr}(\rho H - \rho_E O) + \mathcal{P}_\mu[-\gamma_\mu\psi/2] = 0, \tag{16.4}$$

$$\eta_n^\mu = \sum_i [D_{in}^\mu u_i Y_i + A_{in} r u_i' p_\mu Y_i + A_{in} u_i \mathscr{A}_\mu Y_i], \tag{16.5}$$

$$D^\mu = s^\mu + \tfrac{1}{2}\gamma_\mu A, \tag{16.6}$$

$$\sigma_\mu^{\mathrm{corr}} = \frac{2}{\Omega}\,\mathrm{Re}\sum_{a\mathbf{k}} w_{\mathbf{k}} \sum_{x=0,r,\mathrm{ang}} \mathrm{Tr}_b[\rho Q_x^\dagger (H^x + V^x) A - \rho_E Q_x^\dagger S^x A], \tag{16.7}$$

$$Q_0 = D^\mu, \qquad Q_r = Q_{\mathrm{ang}} = A, \tag{16.8}$$

$$D_\mu^{\mathrm{proj}} = 2\,\mathrm{Re}\sum_{\mathbf{k}} w_{\mathbf{k}}\,\mathrm{Tr}_c[C\partial_\mu U], \tag{16.9}$$

$$U = PX, \qquad \partial_\mu U = (\partial_\mu P)X + P\partial_\mu X, \tag{16.10}$$

$$Y\partial_\mu X + (\partial_\mu X)Y = -X(\partial_\mu \mathcal{S})X, \qquad Y = \mathcal{S}^{1/2}, \tag{16.11}$$

$$UC = \rho_E - \mathcal{E}\rho = B_L, \tag{16.12}$$

$$\mathcal{P}_\mu[b] + D_\mu^{\mathrm{proj,frame}} = 0 \quad \text{for a consistent pure frame change.} \tag{16.13}$$

In the compact Q_x notation, H^0, V^0, S^0 pair with D^μ; the other two channels pair with A. This sheet is a restatement of the derived formulas, not an independent shortcut for the omitted potential, core, ion, boundary, or relativistic terms.

References

[1] Belbase, Tröster, and Blaha, *Stress tensor in the linearized augmented plane wave method*, Physical Review B **104**, 174113 (2021). doi:10.1103/PhysRevB.104.174113. Source inspected: `/Users/haule/papers_to_read/Blahas_stress_tensor.pdf`. Equations (10)–(13), (19)–(20), Appendix D, and Appendix G.

[2] K. Haule, project documentation on the DFT+eDMFT force functional, `/Users/haule/projects/eDMFT/src/doc/Forces_LAPW.pdf`. Background for the stationary-functional and normalized-projector discussion; the strain derivation is given explicitly in this note.

[3] Earlier project working note, `src/doc/stress/edmft_stress_implemented_equations.tex` and `Analytic_Stress_DFT_eDMFT_Implemented_Equations.pdf`, 5 November 2026. The present note supplements that record and makes the Eq. (13) cancellation and local G2 matrix definitions explicit.

[4] Local working tree `/Users/haule/projects/eDMFT_new/`, read-only source inspection, 6 November 2026. The source map in Sec. 14 identifies relevant routines; implementation correctness remains subject to the stated audit.